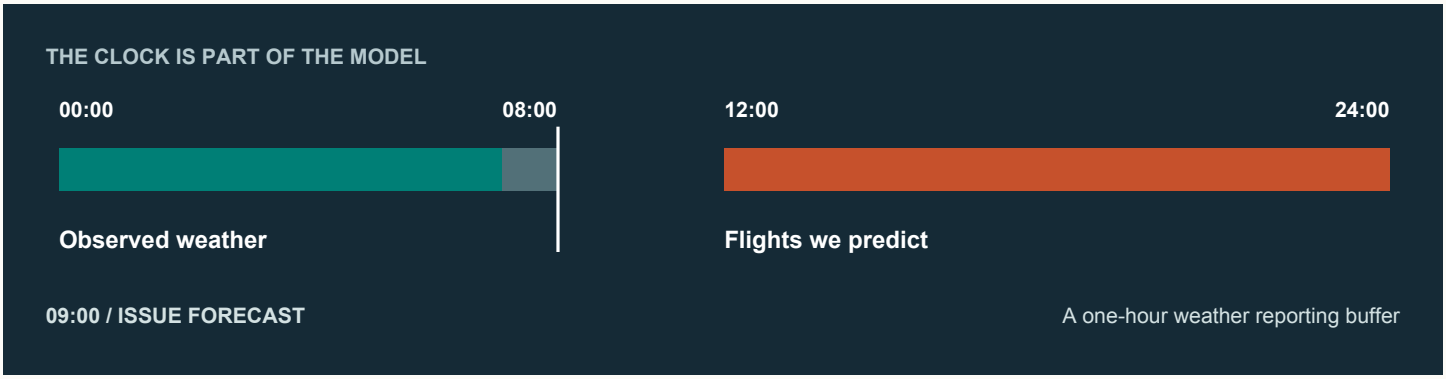
AN AIRPORT-DAY FORECAST, BUILT BEFORE LUNCH

The 9 a.m. forecast

Can morning weather help airport planners anticipate afternoon departure delays? We combine flight schedules with weather recorded before 8 a.m., predict the mean delay of flights scheduled from noon onward, and test the model on a later month. Weather cuts validation error by **12.3%** in 2023. In January 2024, that advantage disappears. A few severe days overwhelm the forecast.



Data

We combine **BTS flight records** with **Iowa Environmental Mesonet weather observations**. The 2023 flights come through nycflights23; January 2024 comes directly from BTS. Both use the same reporting-carrier definitions and contain the same 14 carriers. We aggregate each source by airport and local date, then check a one-to-one join. The panel contains **1,188 airport-days**: 1,095 for development and 93 for the future test.

The outcome includes early departures and averages only recorded delays. Of 262,528 scheduled afternoon flights in development, **8,180 lack a delay outcome**; they remain in schedule counts. Development mean delay is **21.1 minutes**, versus a median of **12.8**. Weather is available for every airport-day; missing individual readings use training-only medians. Trace rain remains a small positive amount. No afternoon weather enters the forecast.

1,095
DEVELOPMENT AIRPORT-DAYS

93
FUTURE AIRPORT-DAYS

08:00
WEATHER CUTOFF

Schedules are reconstructed from final records, so historical schedule revisions and feed latency remain untested. Missing delay outcomes, including cancellations, are a separate operational risk. January supplies 7.8% of the combined panel.

Sources: BTS / nycflights23 / Iowa Environmental Mesonet

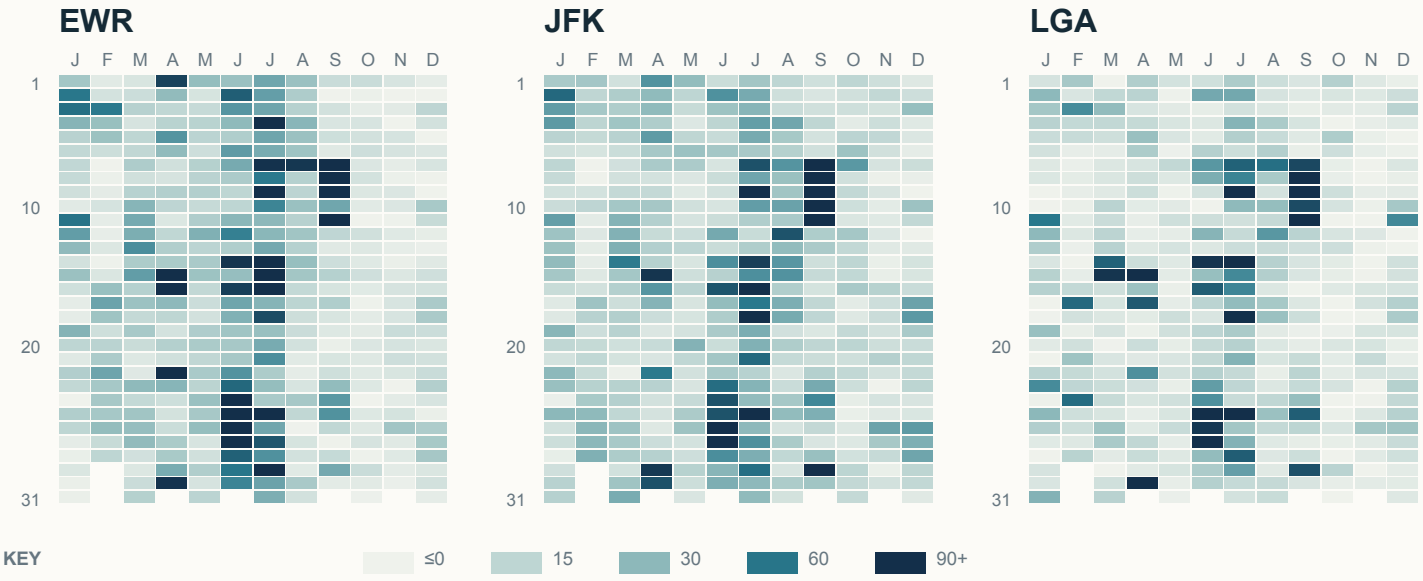
01 / SEASON AND WEATHER

Bad mornings are not the whole story.

Delay clusters in summer, including days without low visibility or strong morning wind. Weather matters differently across airports and seasons.

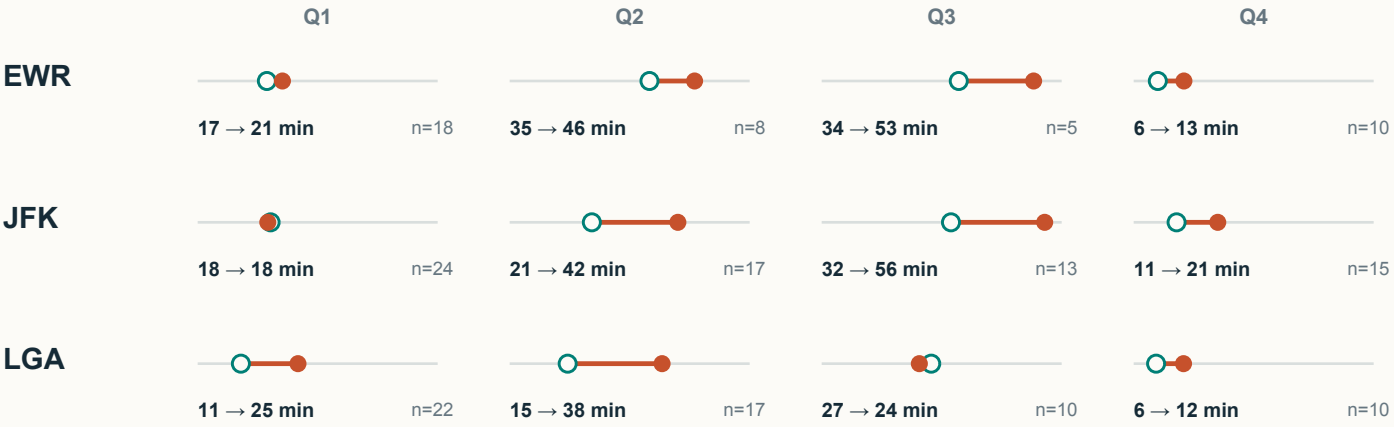
Every afternoon in 2023

Mean departure delay, minutes



Within the same airport and quarter

Mean afternoon delay: other mornings ○ → low visibility / high wind ●



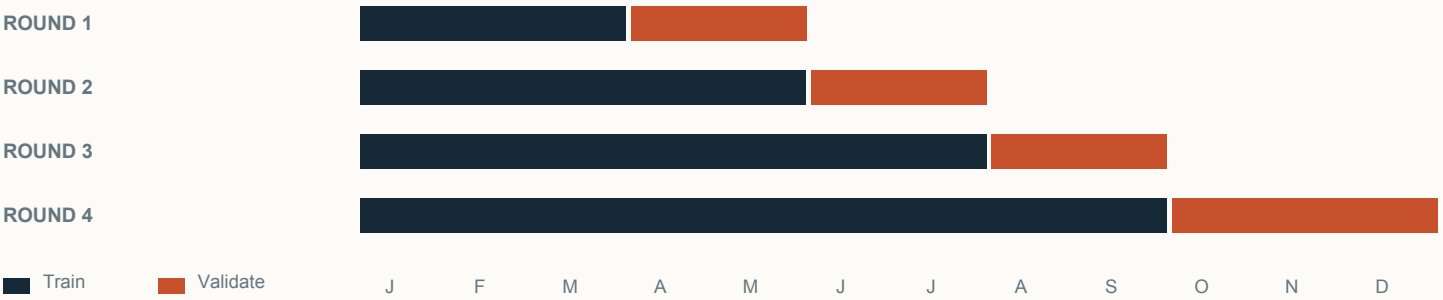
Weather flag: minimum pre-8 a.m. visibility below 3 miles or maximum wind at least 20 knots. Counts show flagged mornings; some cells are small. Mean delay is higher in 10 of 12 comparisons, not all 12. These are descriptive associations, not causal estimates.

02 / PAIRED FORWARD COMPARISONS

Give weather a fair test.

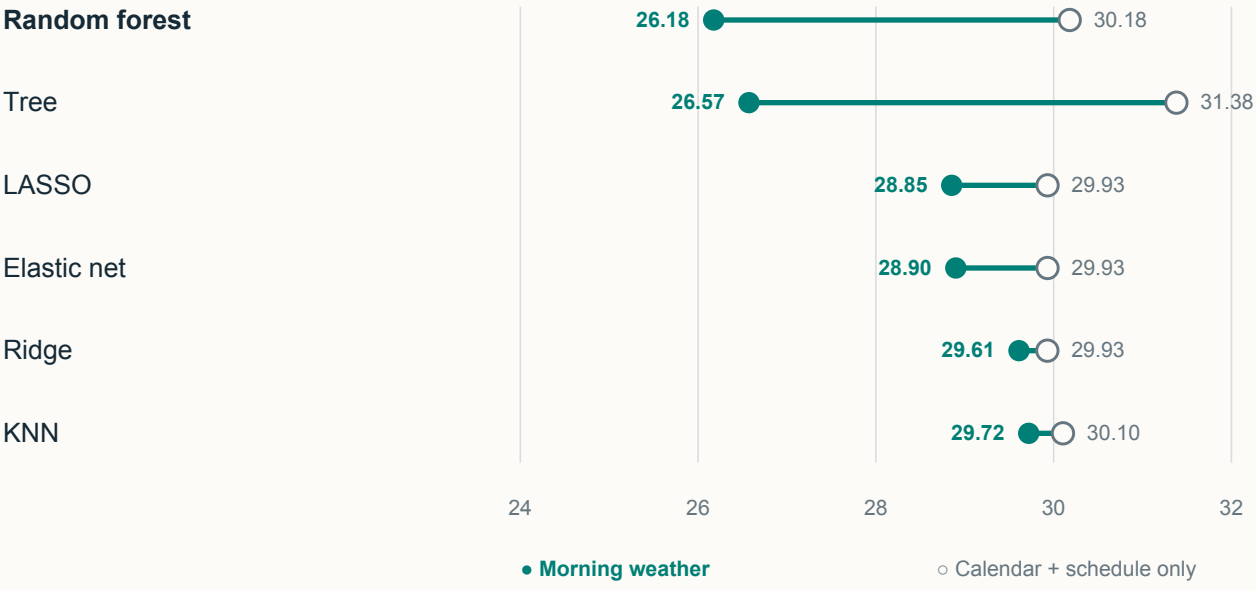
Tune each method twice: once with calendar and schedule inputs, then with those inputs plus morning weather. Both versions predict the same later dates.

The validation clock only moves forward



Weather helps more than changing the algorithm.

Pooled forward-validation RMSE, minutes • lower is better



The strongest no-weather choice is the airport average: **29.84 minutes RMSE**. The weather forest lowers that to **26.18**, or **12.3%**. But the gain varies by season:

APR-MAY	JUN-JUL	AUG-SEP	OCT-DEC
24.8 → 23.4	44.9 → 34.0	28.7 → 33.1	19.2 → 13.8

Airport-average → morning-weather RMSE, minutes. Each window is later than its training data.

03 / SELECT BEFORE TESTING

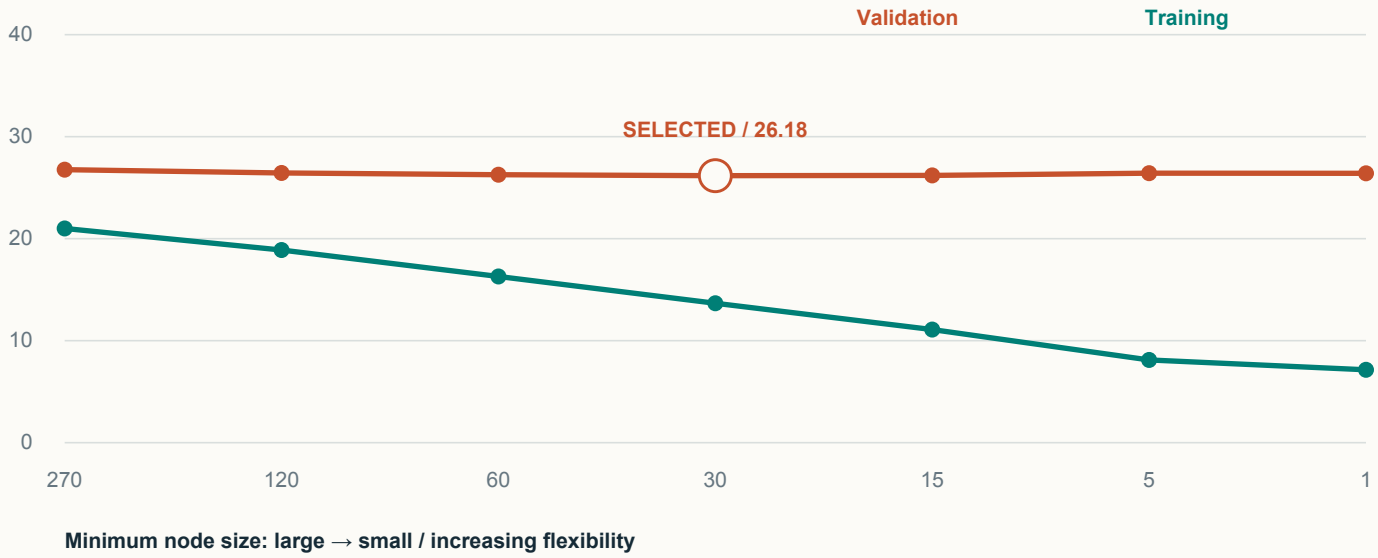
A restrained forest wins.

The selected model averages 500 trees, considers all 22 predictors at a split, and stops splitting nodes smaller than 30 observations. All tuning uses 2023 only.

Inputs cover airport, weekday, annual seasonality, afternoon schedule size, route distance, long-haul and evening shares, carrier mix, and early weather. Penalized regressions also receive quadratic terms, numeric interactions and airport interactions. Imputation and scaling are fitted separately inside each training fold. There are **828 candidate configurations** across the two information sets, plus four fixed baselines.

Training fit improves. Future-like validation does not.

Forest with weather • all 22 predictors available at each split



The search did not stop at depth 6.

Tree: 18 depths, up to 30, crossed with four leaf sizes. The best weather tree uses depth 3.

Forest: four predictor counts crossed with seven node sizes, including very shallow trees.

Ridge / LASSO / elastic net: 61 penalties from 0.001 to 1,000,000 and five mixing values.

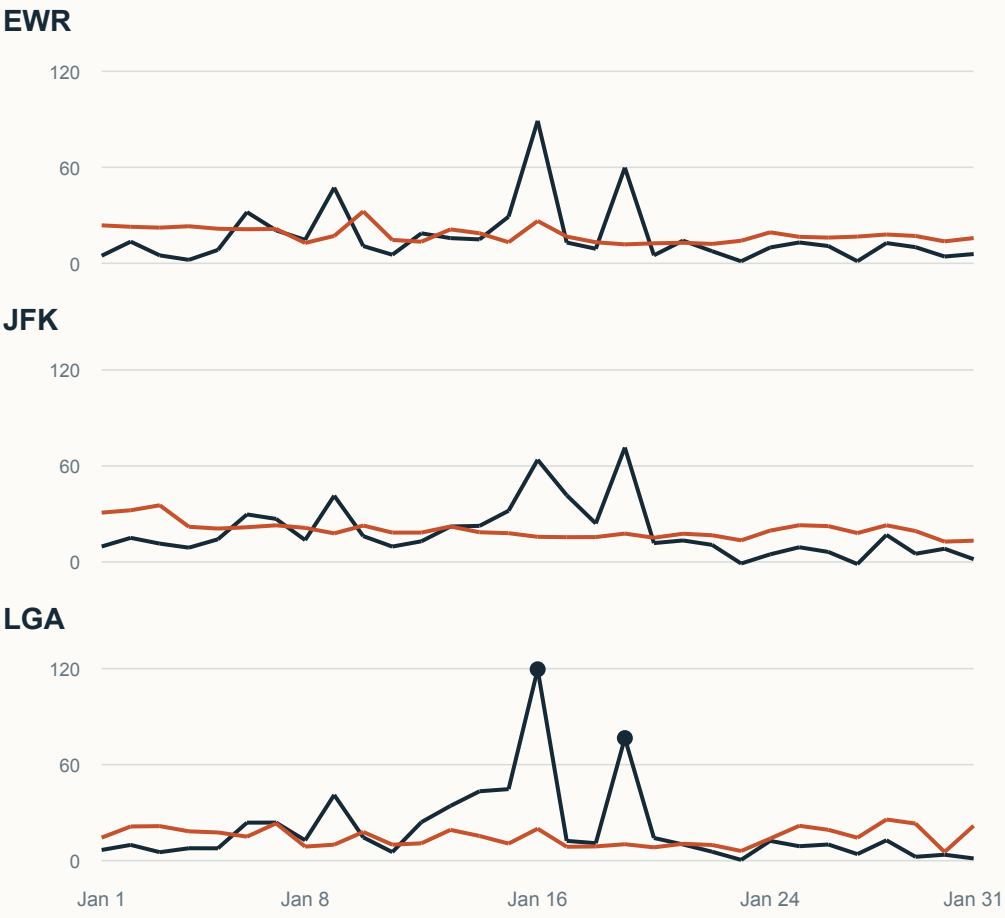
KNN: nine neighbor counts, from 3 to 240. Boundary optima triggered wider grids before evaluation.

The January 2024 outcomes played no role in choosing the method, settings or benchmark. All models were fitted on the full 2023 development sample, and their future predictions were saved before evaluation outcomes were joined.

04 / JANUARY 2024 · 31 DATES · 93 AIRPORT-DAYS

January breaks the winning streak.

Morning weather does not lower future RMSE. Against the airport average, MAE falls from 15.12 to 13.98 minutes, but the largest delays go largely unanticipated. Three dates account for 69.5% of its squared error.



The same difficult dates hit multiple airports. Uncertainty must keep those shared shocks together.

JAN 16 / LGA

120 min

Predicted: 20 min

20.76
AIRPORT-MEAN RMSE

20.89
WEATHER-FOREST RMSE

13.98
WEATHER-FOREST MAE

No demonstrated RMSE gain. The change is **-0.14 minutes** of improvement. A paired seven-day block bootstrap gives a 95% interval of **-1.44 to +1.70 minutes**. The fixed-family forest comparison also straddles zero.

Decision: do not replace the simple benchmark on this evidence. Test more seasons and add forecasts of afternoon weather and network disruption. January has only 31 dates; its 1,025 missing departure outcomes are another blind spot.

Intervals use 2,000 resamples of seven-day blocks, keeping all airports together and the fitted models fixed. RMSE is in minutes of airport-day mean delay, not individual passenger delay. Lower is better.